

Phase Correlation Analysis of a Gain-Switched DFB Laser

Effect of repetition rate on pulse-to-pulse intensity and phase coherence
with and without CW SLD injection (1 – 10 GHz, step 1 GHz)

DFB Laser (1550 nm, InGaAsP/InP)

Wavelength	1550 nm
Cavity length	300 μm
Threshold current	16.67 mA
Photon lifetime τ_p	3.10 ps
Henry α factor	3.0
Diff. gain a	2.50e-20 m ²
Gain compression ϵ	3.00e-23 m ³
Spont. coupling β	1.0e-04

SLD Injection (CW, broadband)

SLD wavelength	1550 nm
SLD current	150 mA
SLD output power	6.69 mW
Coupling efficiency η	0.10
Injection rate κ	1.114e+10 s ⁻¹
Injected density S_{inj}	4.310e+17 m ⁻³
Frequency detuning $\Delta\nu$	0 GHz

Simulation Parameters

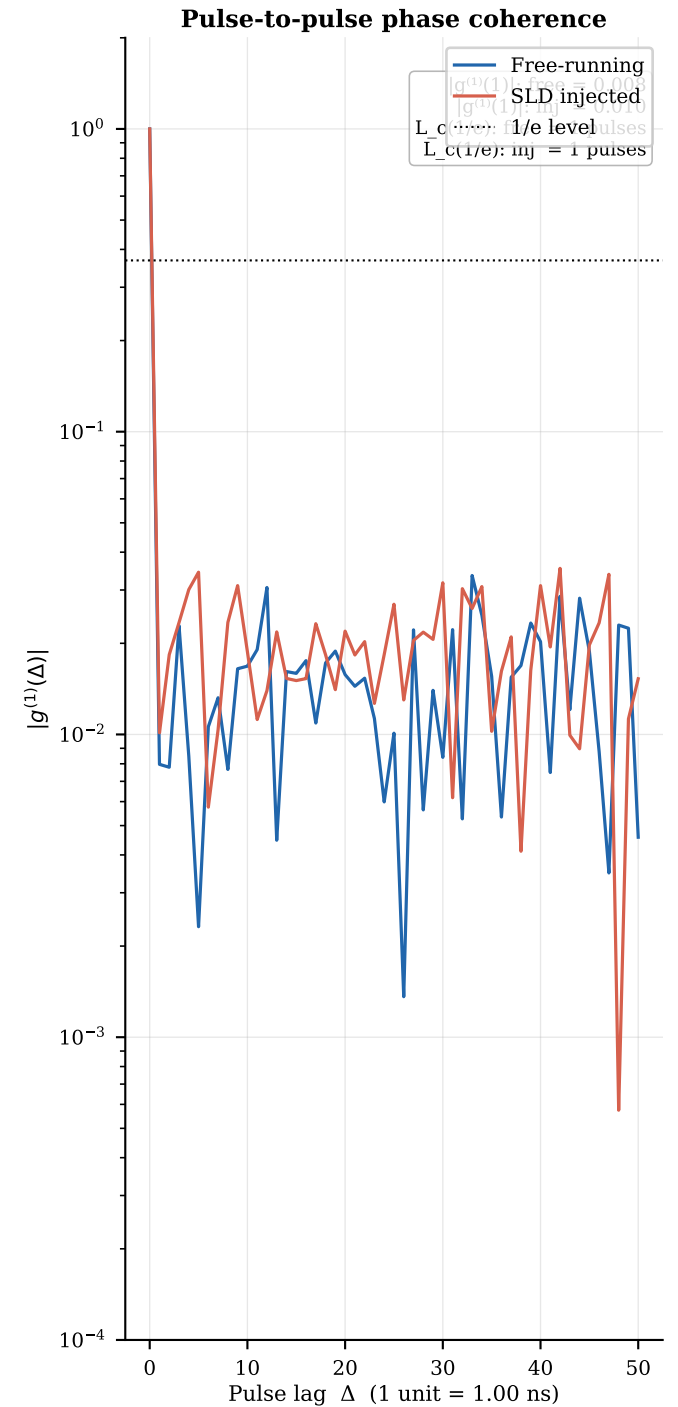
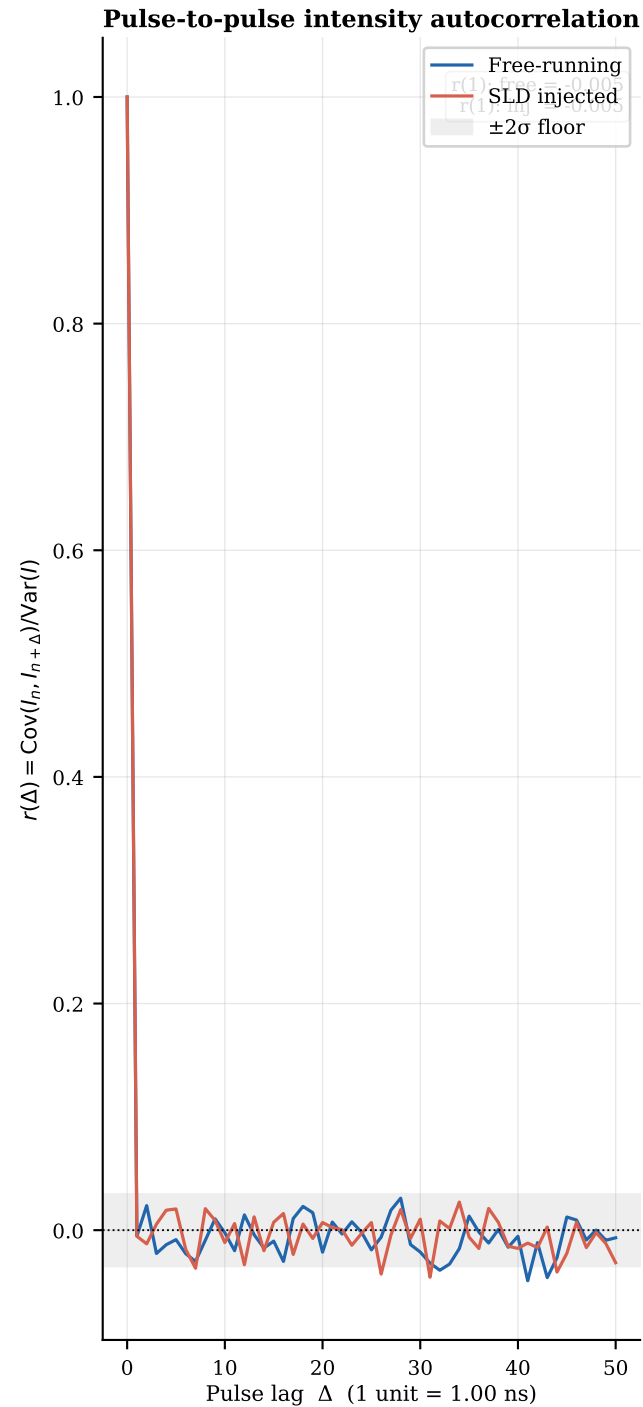
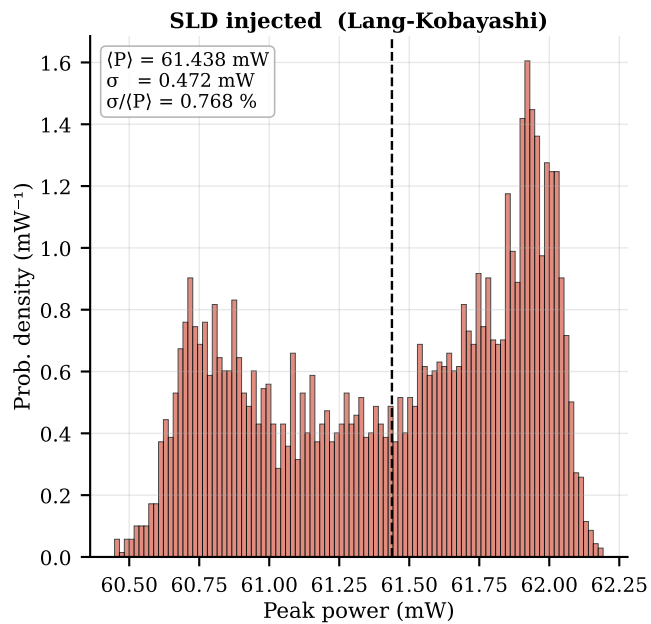
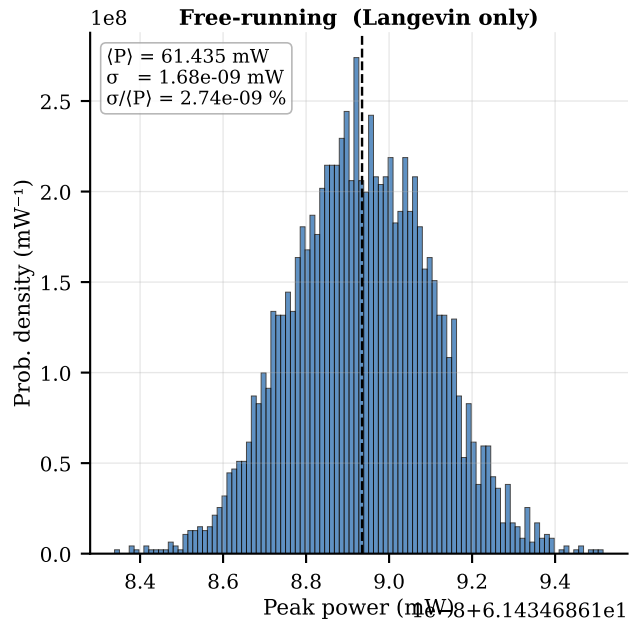
Integration method	Euler-Maruyama (SDE)
Time step dt	2 ps
Parallel trains K	20
Pulses per train	200
Total pulses	4,000
Bias I_{bias}	0.9 × I_{th}
Peak I_{peak}	5.0 × I_{th}

Noise (Langevin / Gardiner)

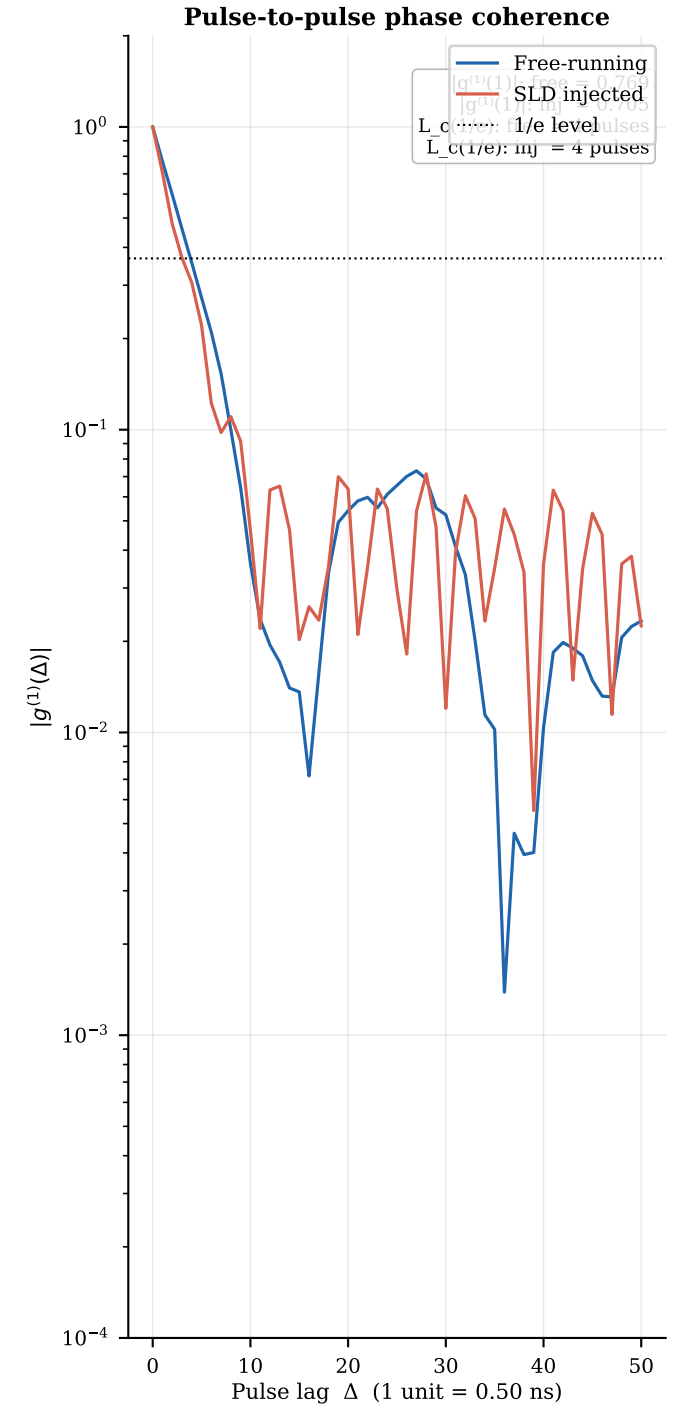
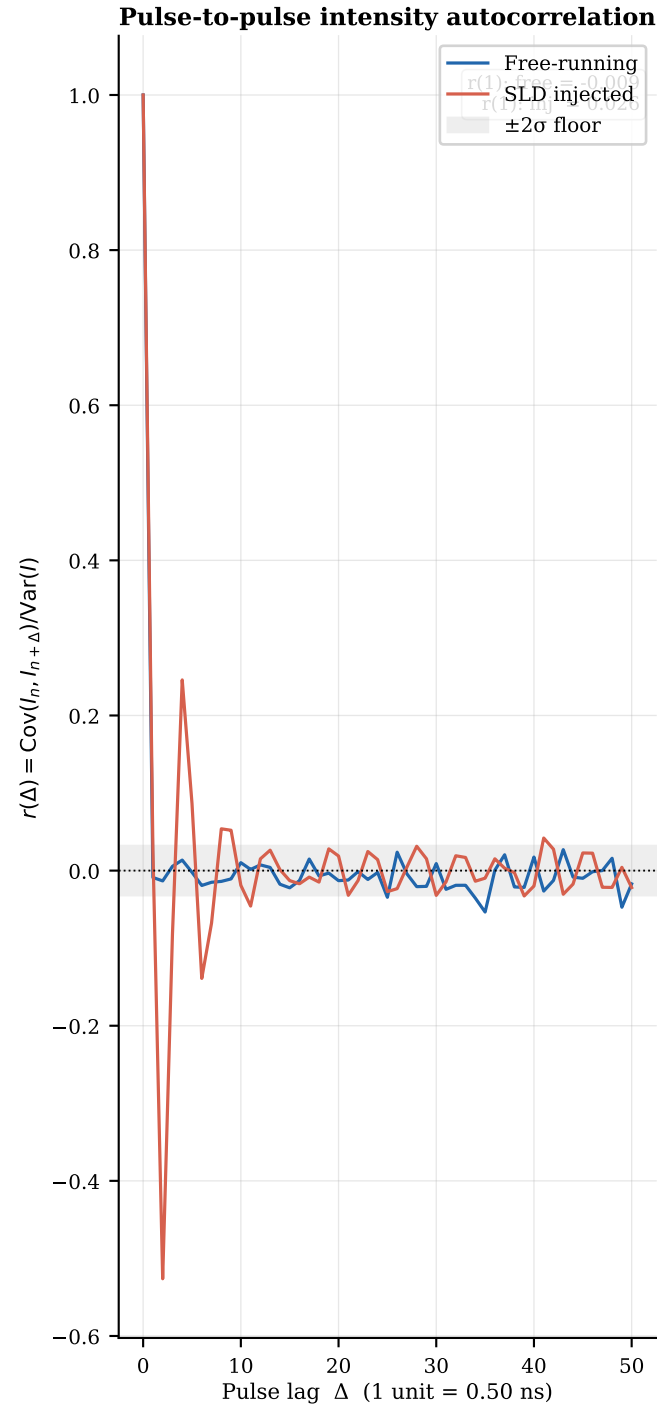
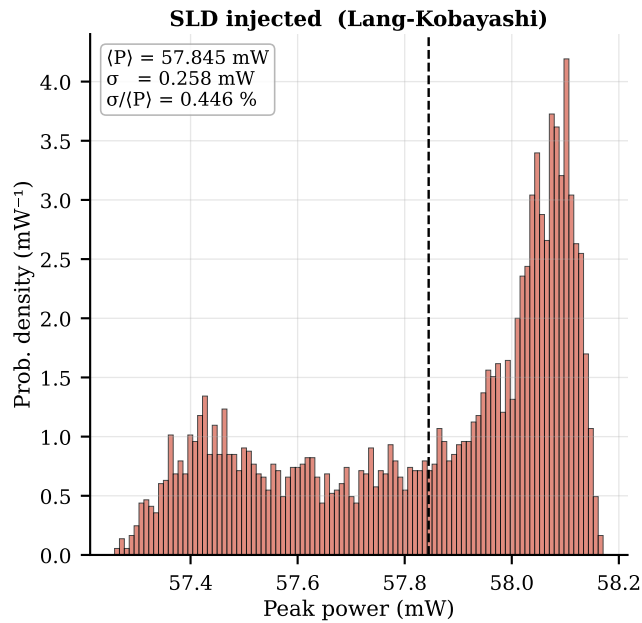
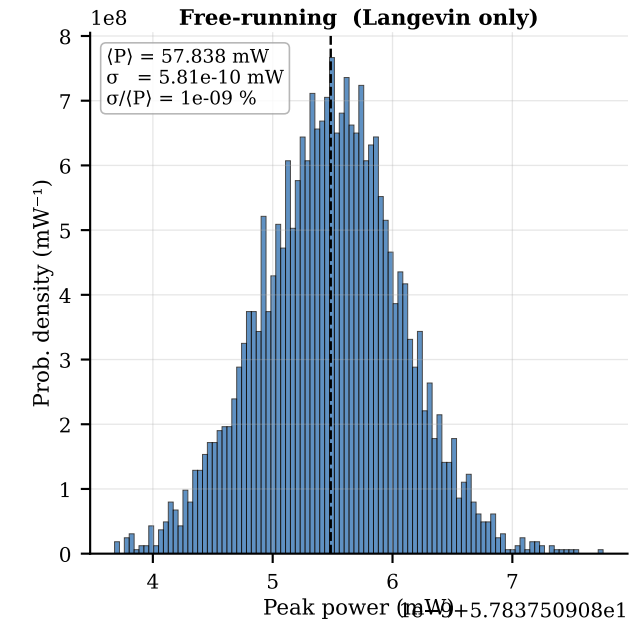
Carrier noise σ_N	$\sqrt{(2 R_{sp})}$ per $\sqrt{s}$
Photon noise σ_S	$\sqrt{(2 \beta B N^2)}$ per $\sqrt{s}$
Phase noise σ_φ	$\sqrt{(\beta B N^2 / 2S)}$ per $\sqrt{s}$
Injection (free-run)	$\kappa = 0, S_{inj} = 0$
Injection (seeded)	κ as above, S_{inj} as above
Lang-Kobayashi term	$2\kappa\sqrt{(S \cdot S_{inj})} \cos \varphi$ in dS/dt

Sampling: photon density S and accumulated phase φ are recorded at the pulse-intensity peak (located via a deterministic burn-in) for each of the 1,000,000 simulated pulses.
Rise time t_{rise} 5% of T_{rep}

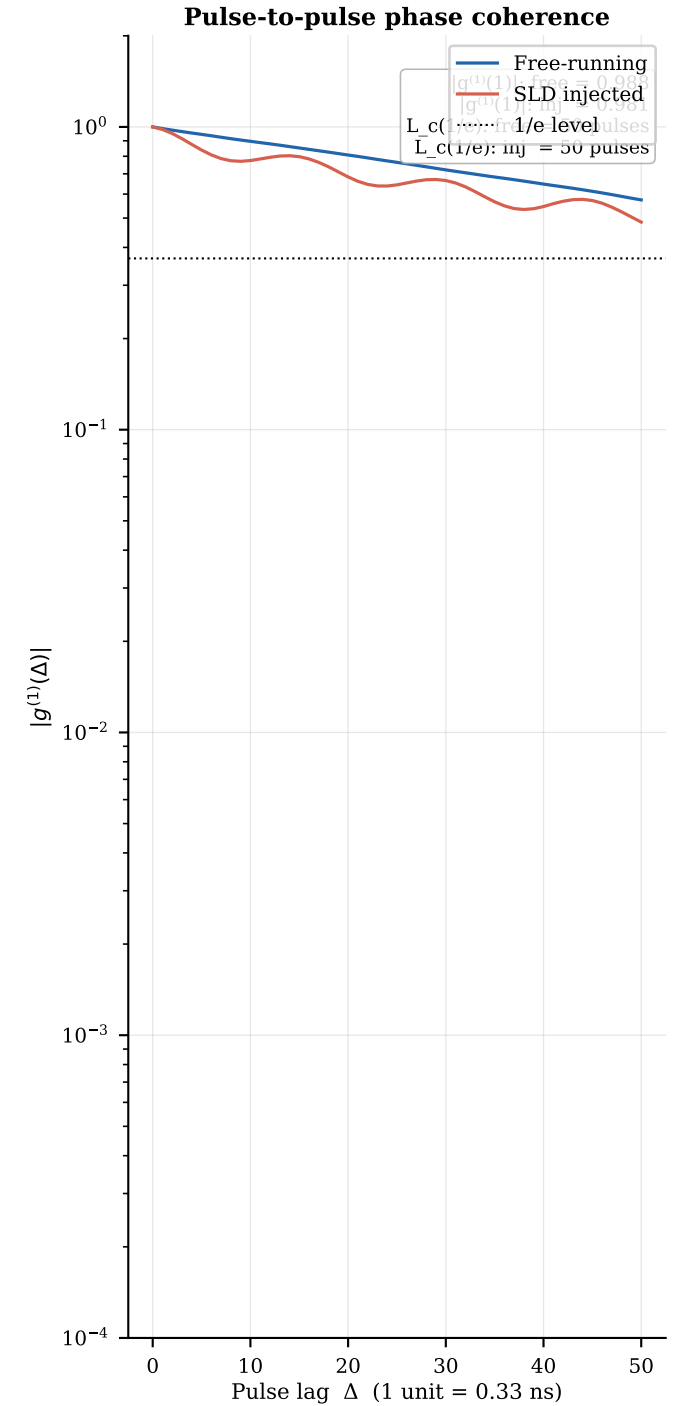
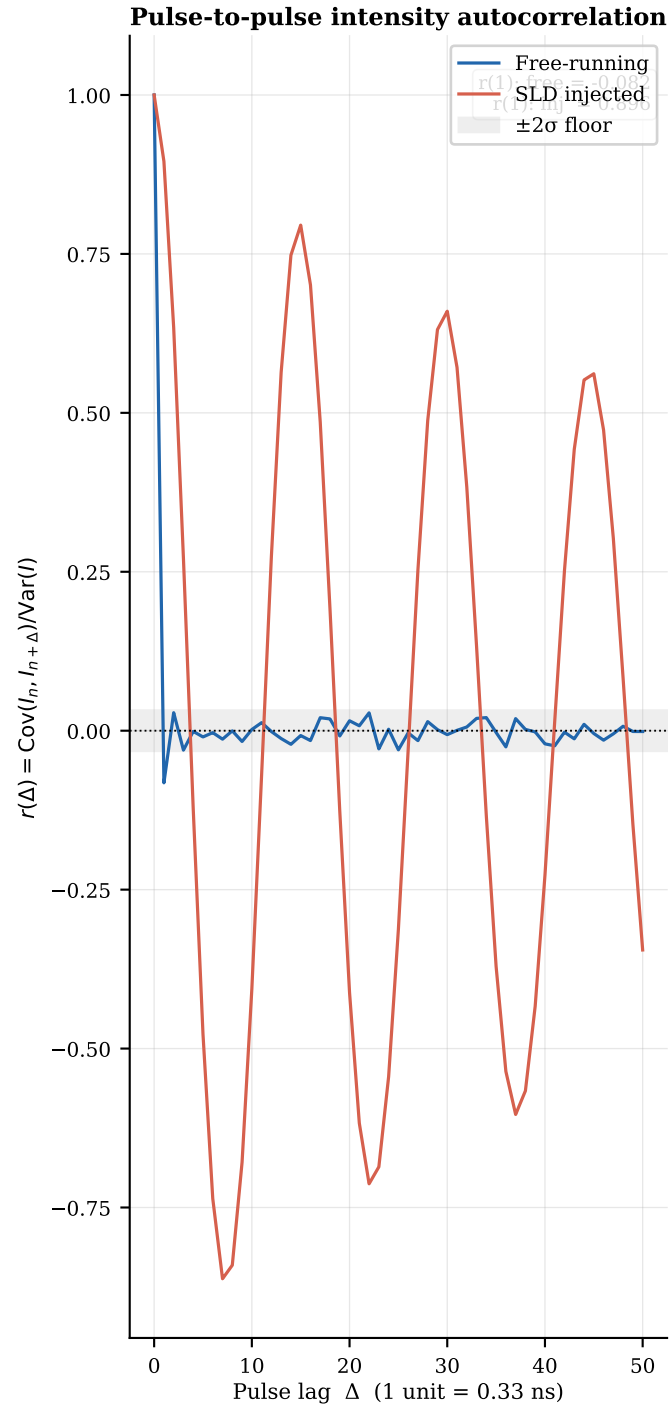
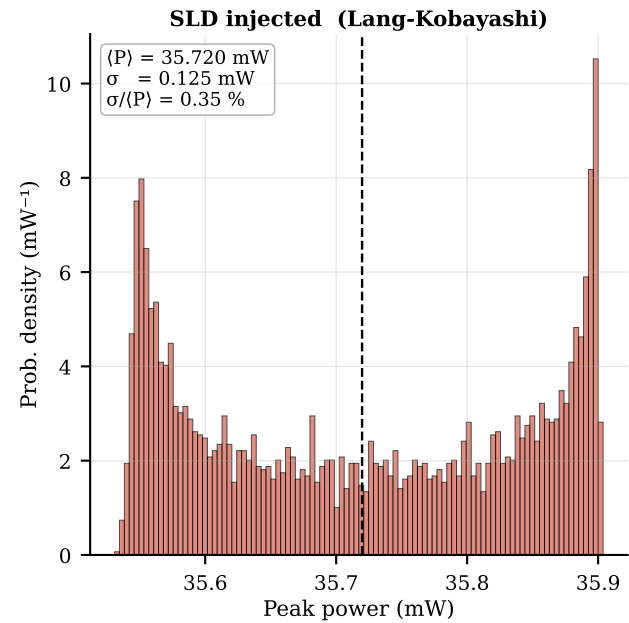
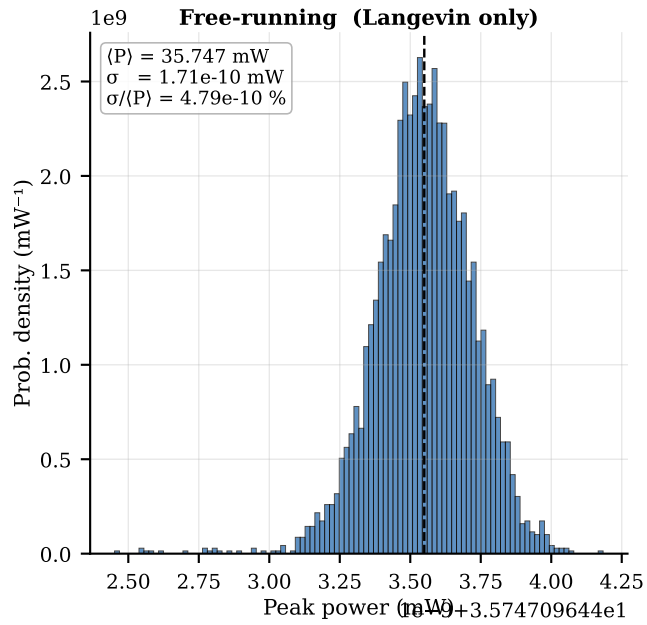
Repetition rate $f_{\text{rep}} = 1 \text{ GHz}$ ($T_{\text{rep}} = 1.00 \text{ ns}$) $N = 4,000 \text{ pulses}$ sampled at $t = 162.0 \text{ ps}$ into each pulse



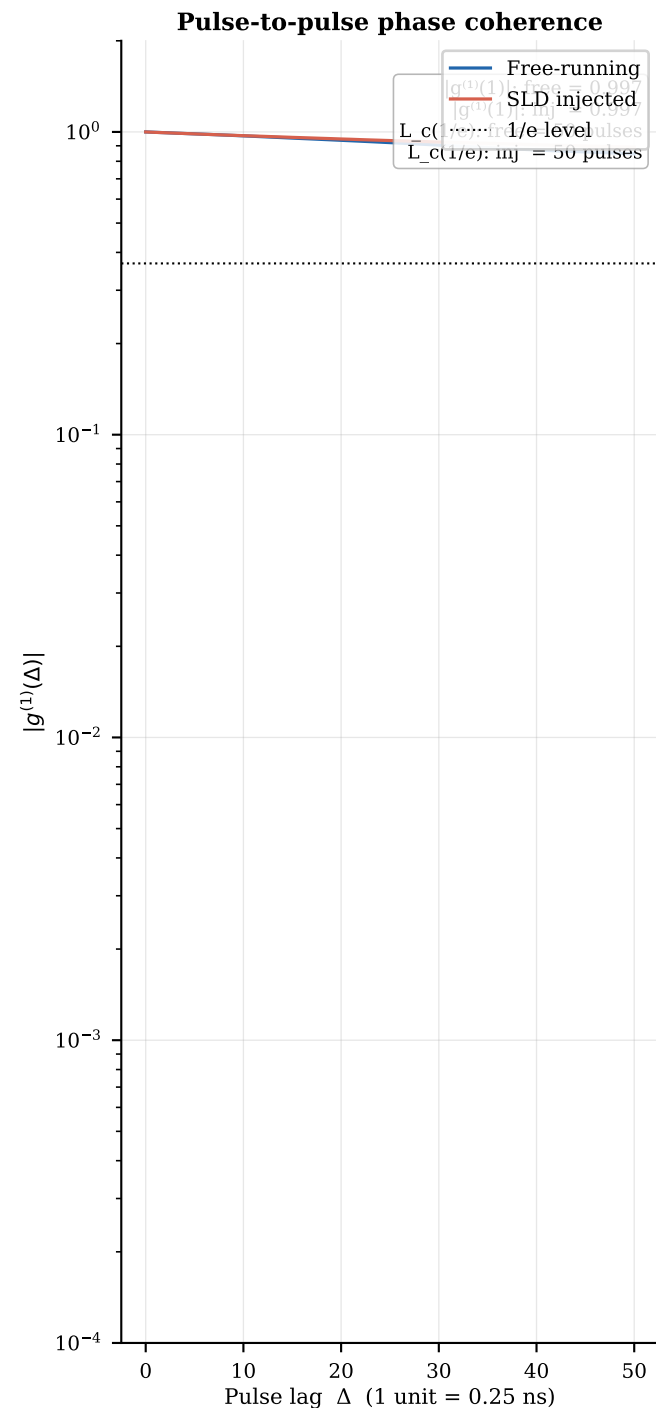
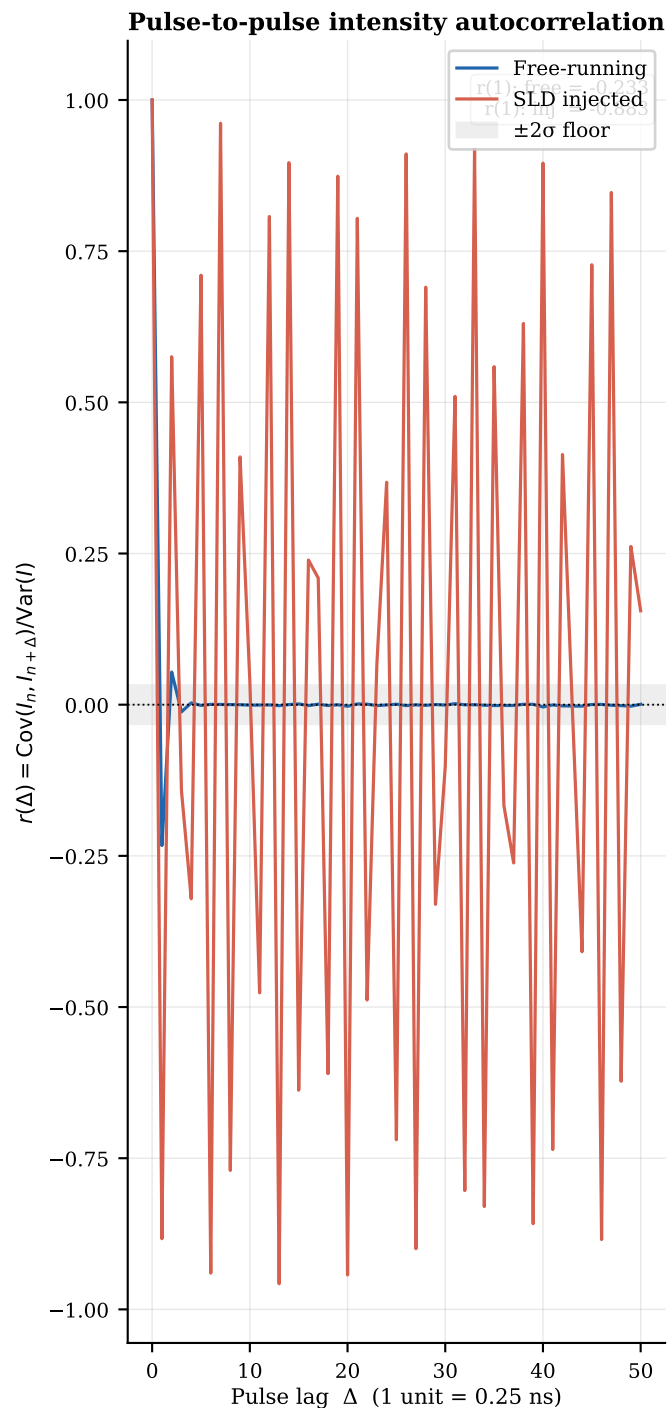
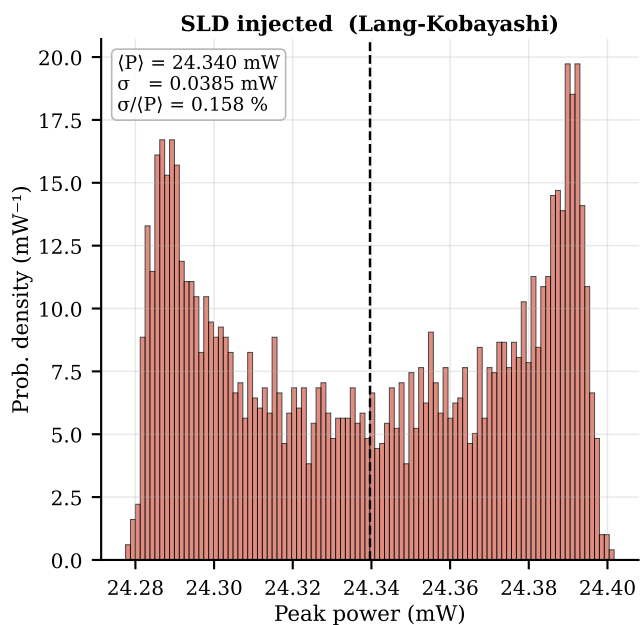
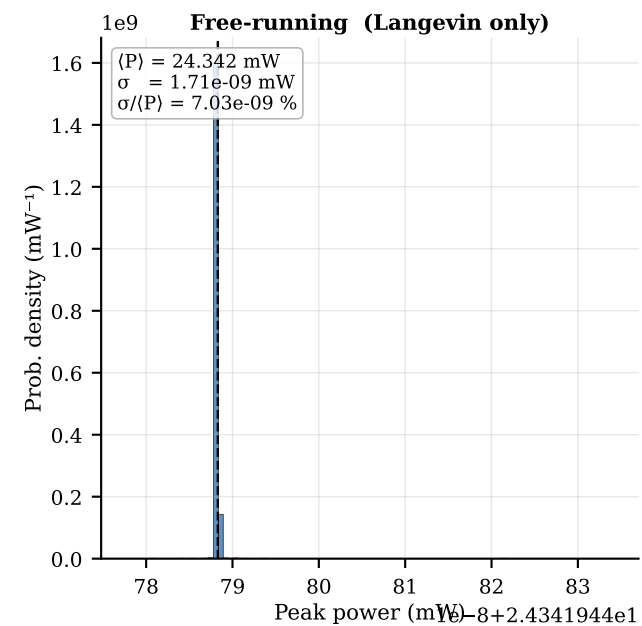
Repetition rate $f_{\text{rep}} = 2 \text{ GHz}$ ($T_{\text{rep}} = 0.50 \text{ ns}$) $N = 4,000 \text{ pulses}$ sampled at $t = 140.0 \text{ ps}$ into each pulse



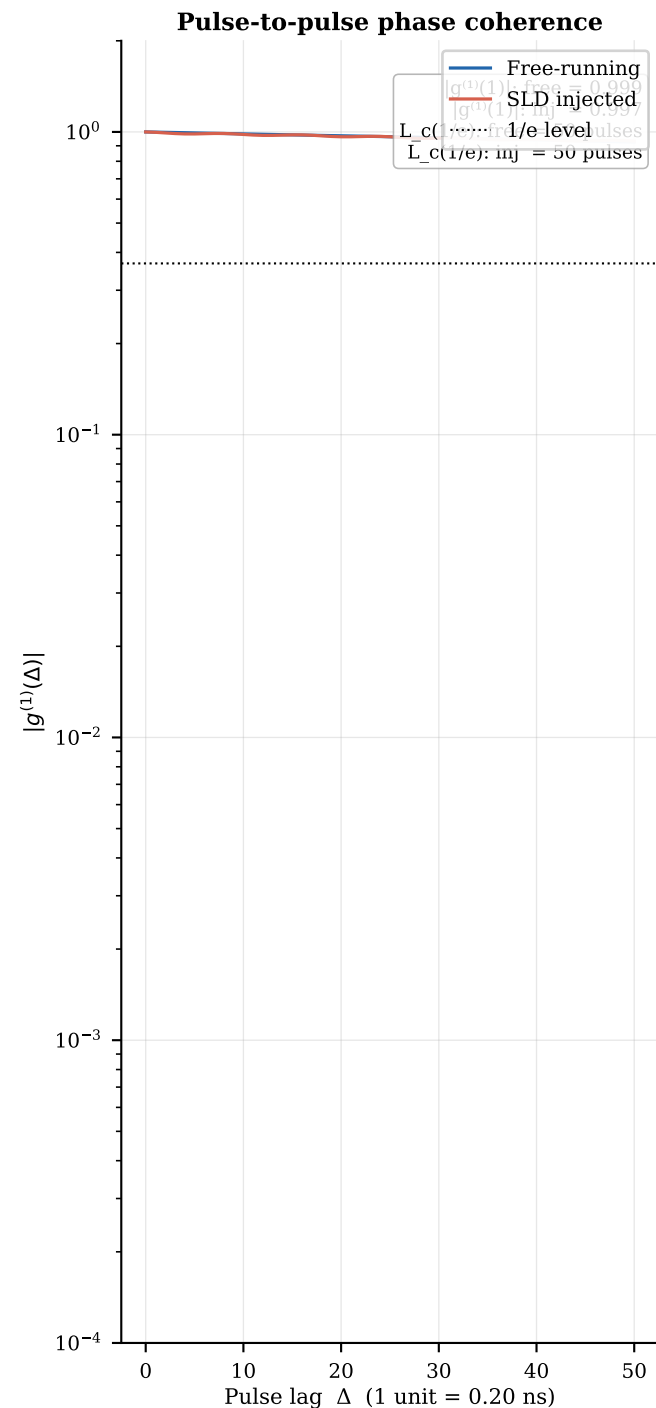
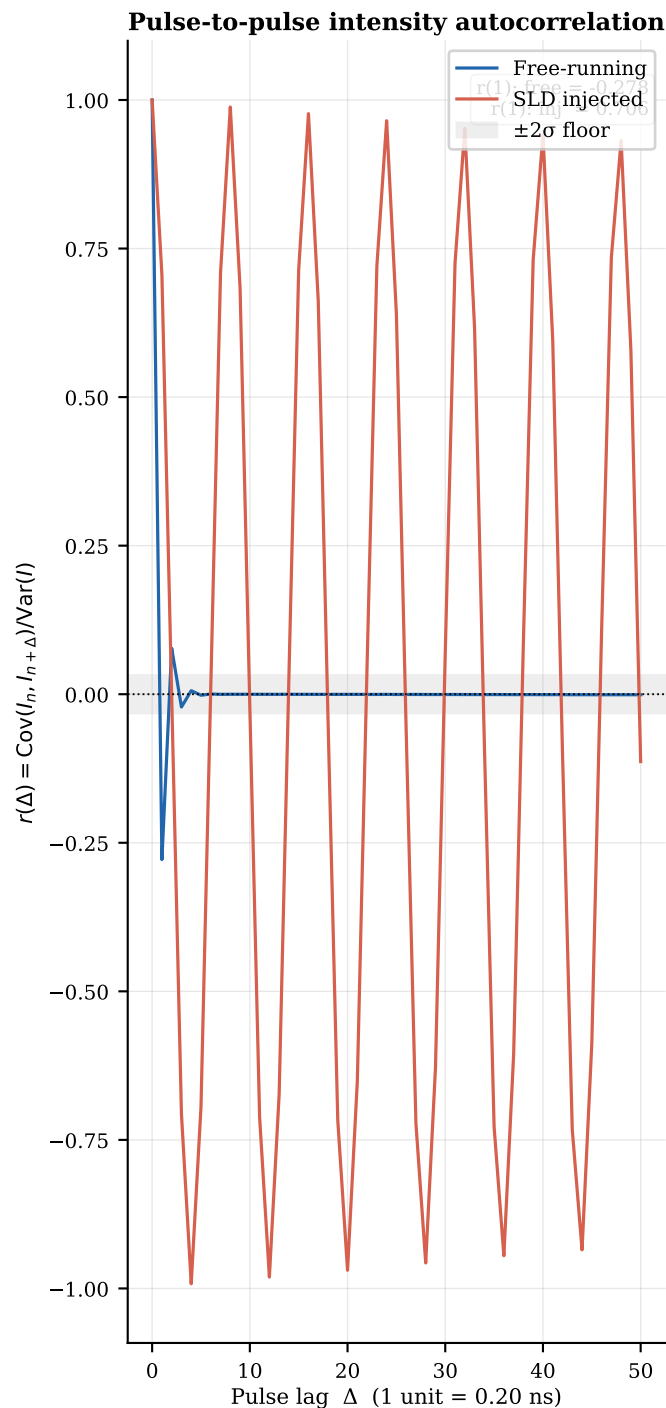
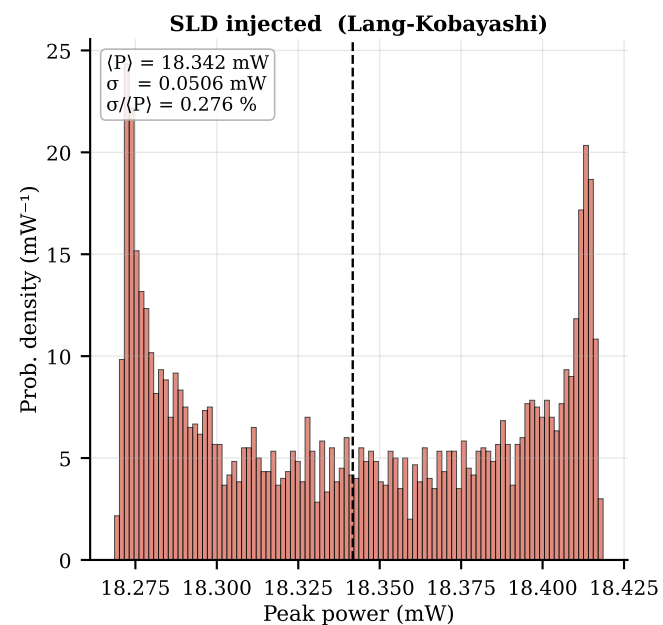
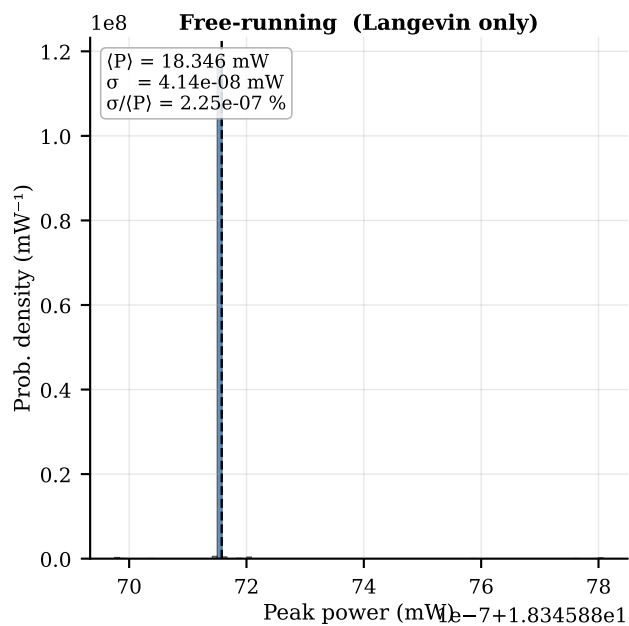
Repetition rate $f_{\text{rep}} = 3 \text{ GHz}$ ($T_{\text{rep}} = 0.33 \text{ ns}$) $N = 4,000 \text{ pulses}$ sampled at $t = 112.0 \text{ ps}$ into each pulse



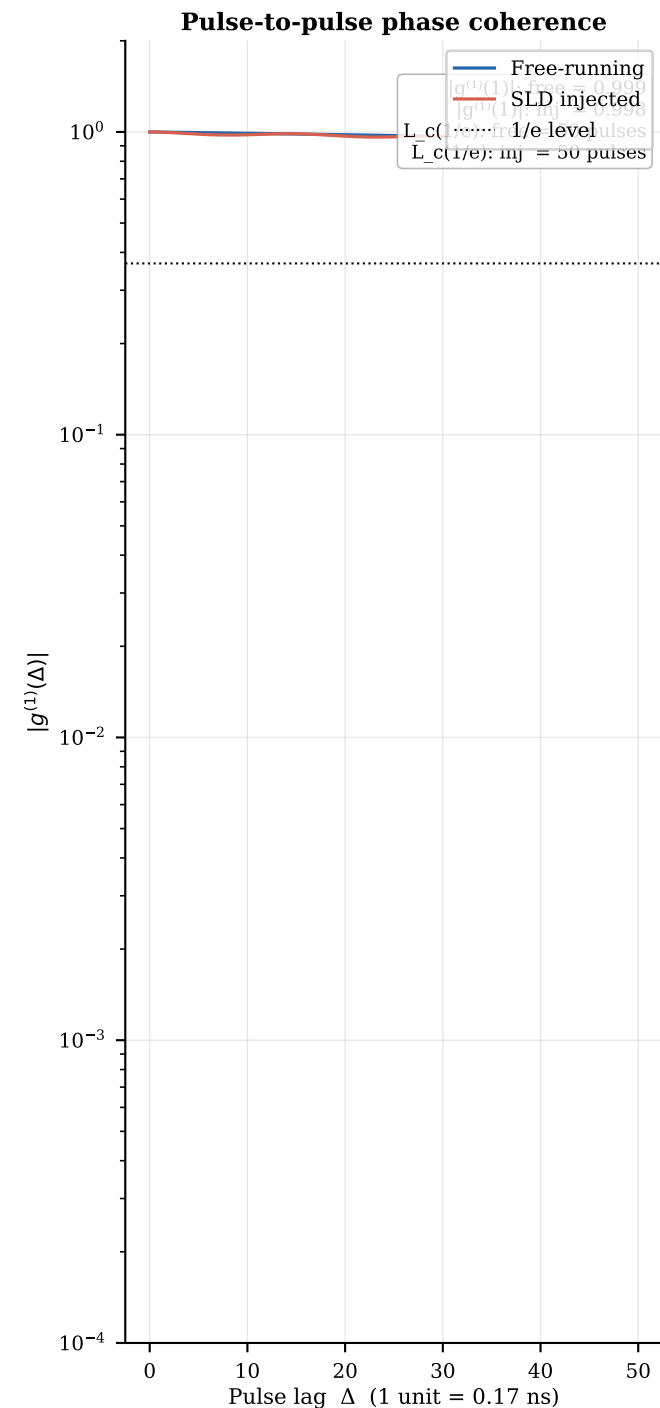
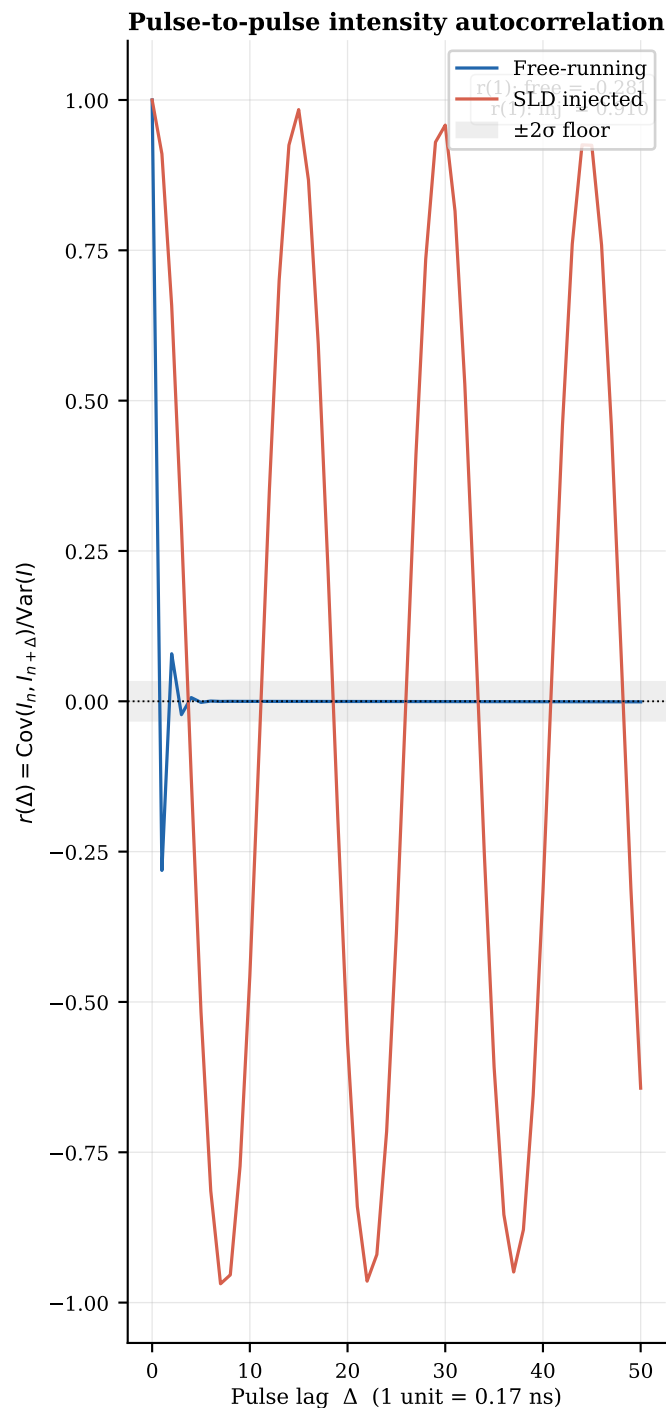
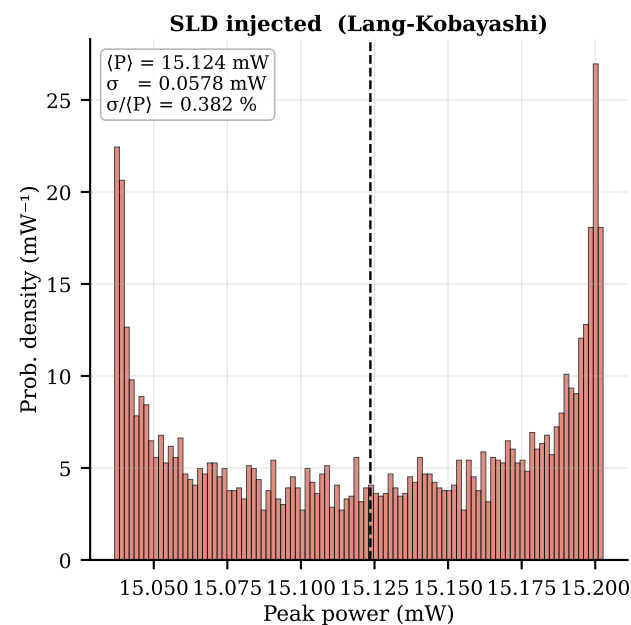
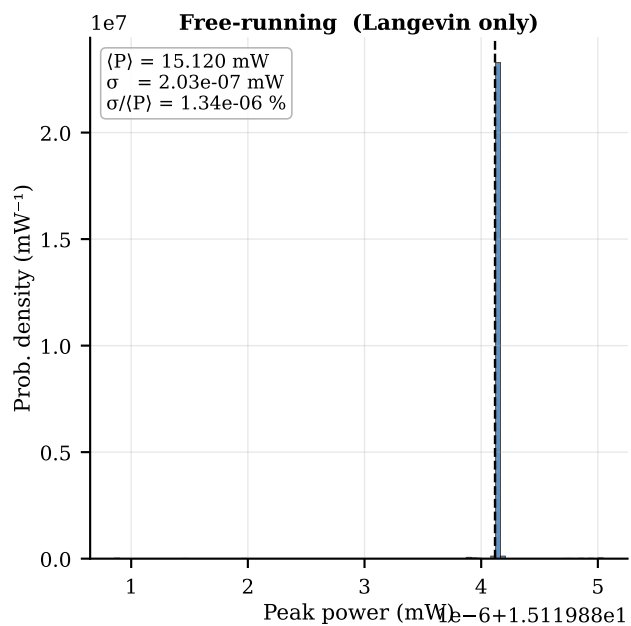
Repetition rate $f_{\text{rep}} = 4 \text{ GHz}$ ($T_{\text{rep}} = 0.25 \text{ ns}$) $N = 4,000$ pulses sampled at $t = 98.0 \text{ ps}$ into each pulse



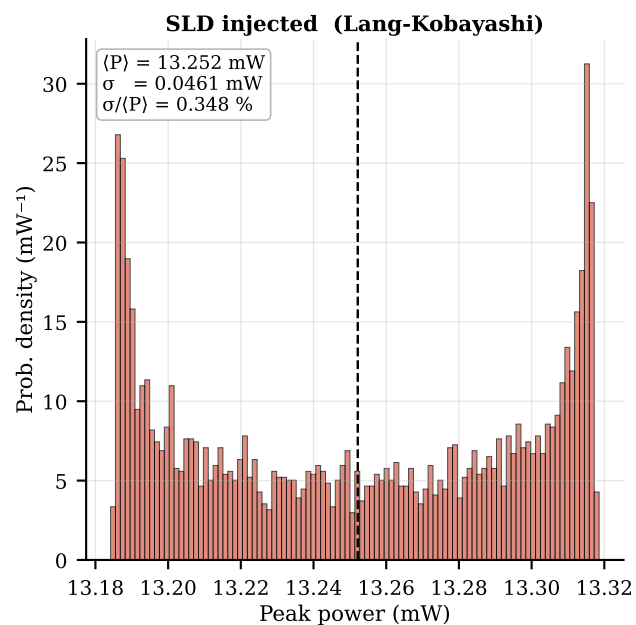
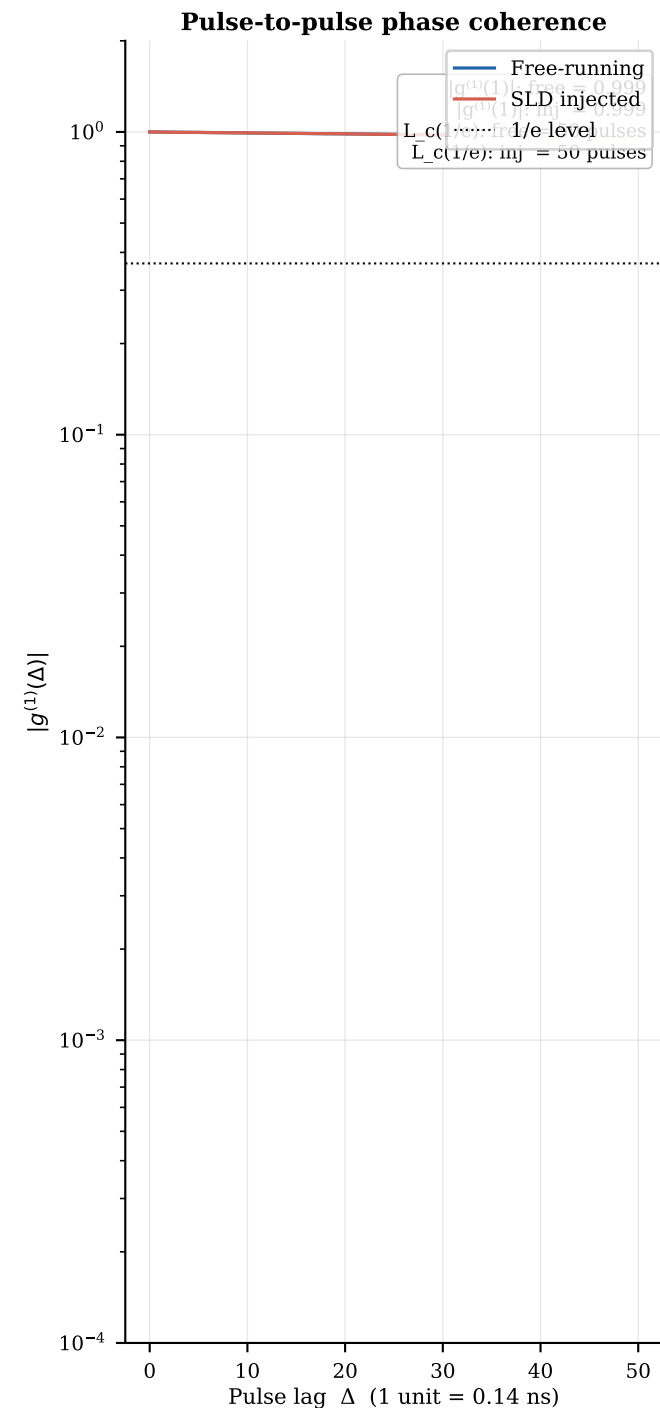
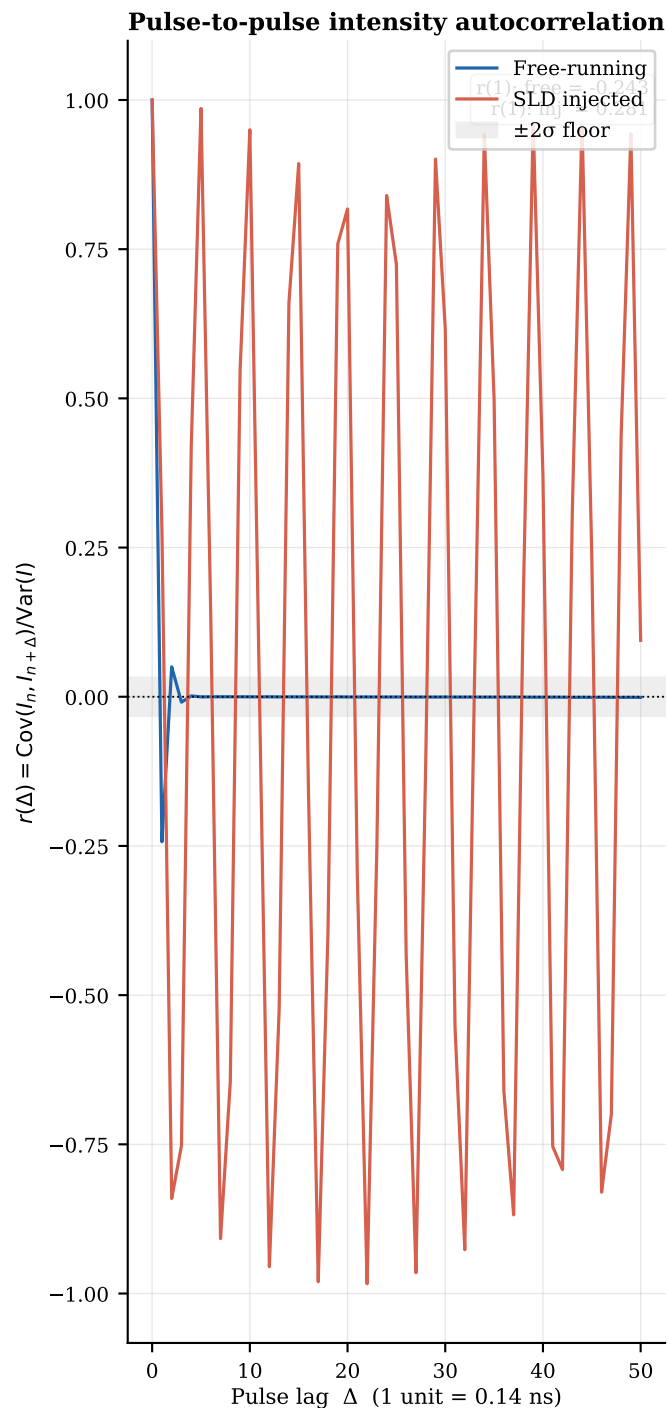
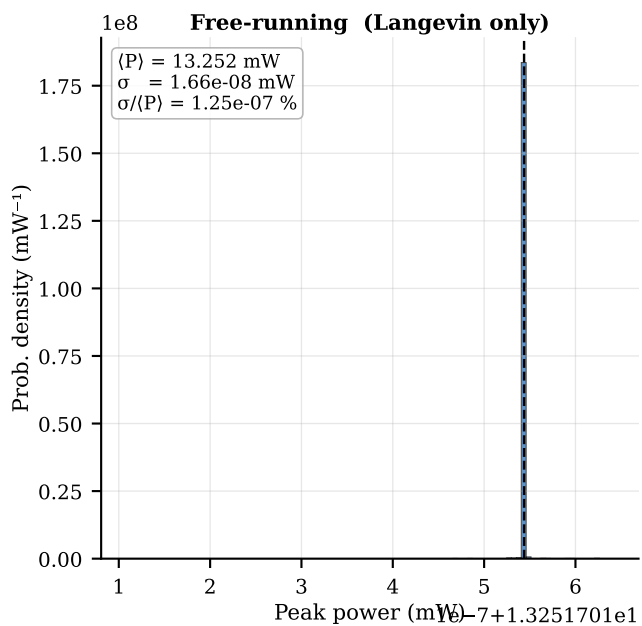
Repetition rate $f_{\text{rep}} = 5 \text{ GHz}$ ($T_{\text{rep}} = 0.20 \text{ ns}$) $N = 4,000$ pulses sampled at $t = 88.0 \text{ ps}$ into each pulse



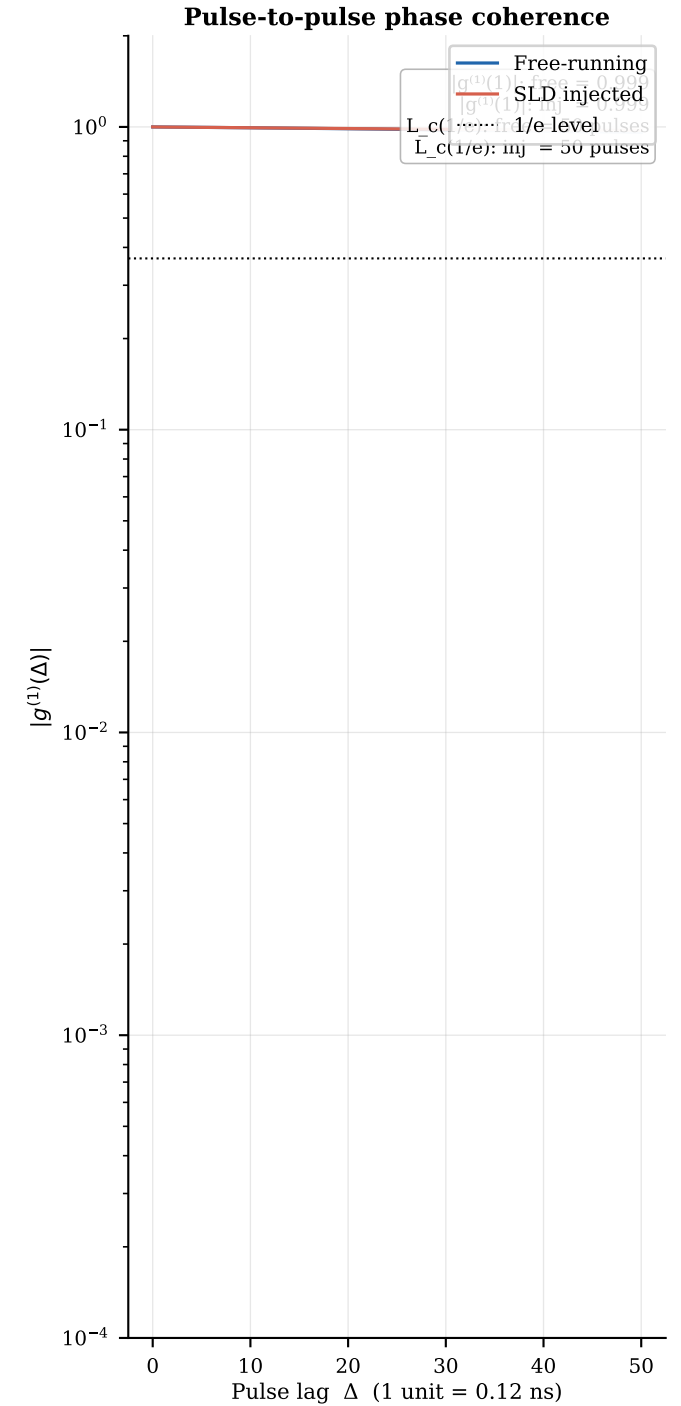
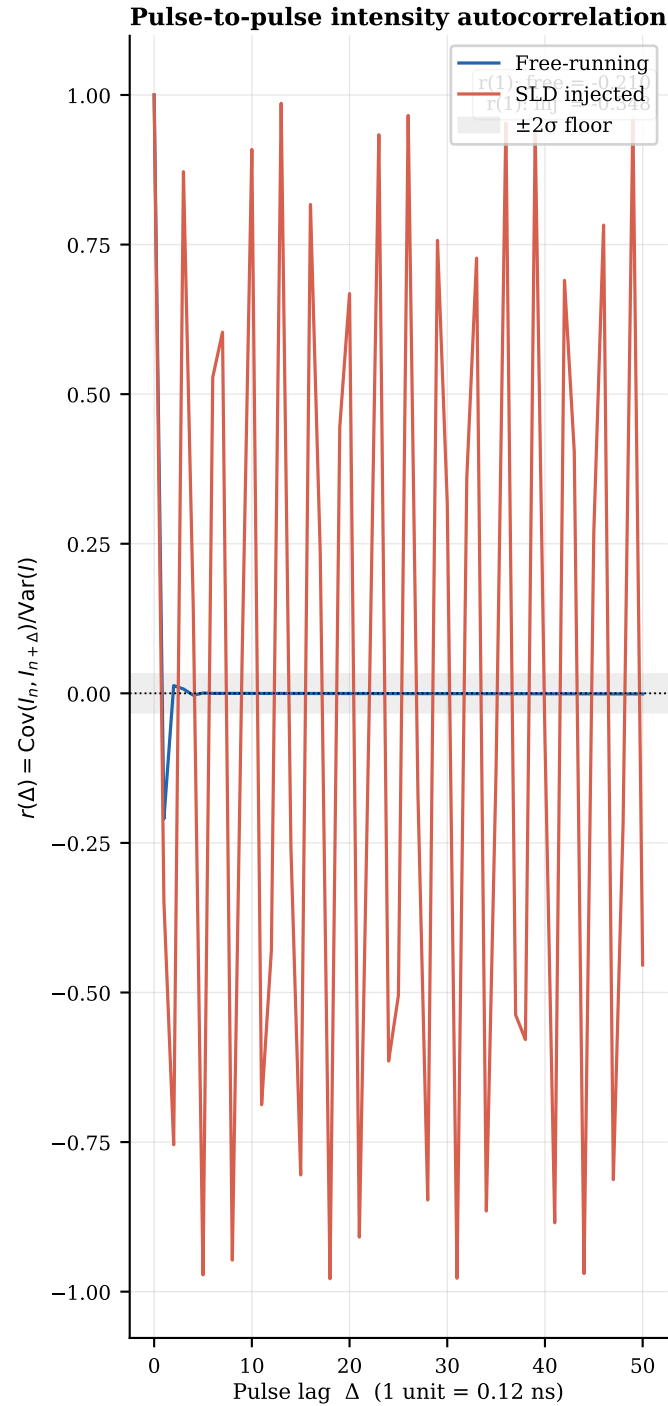
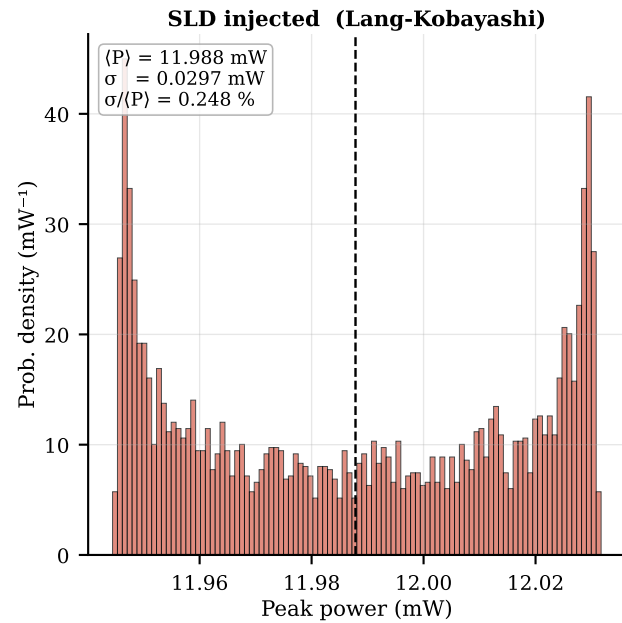
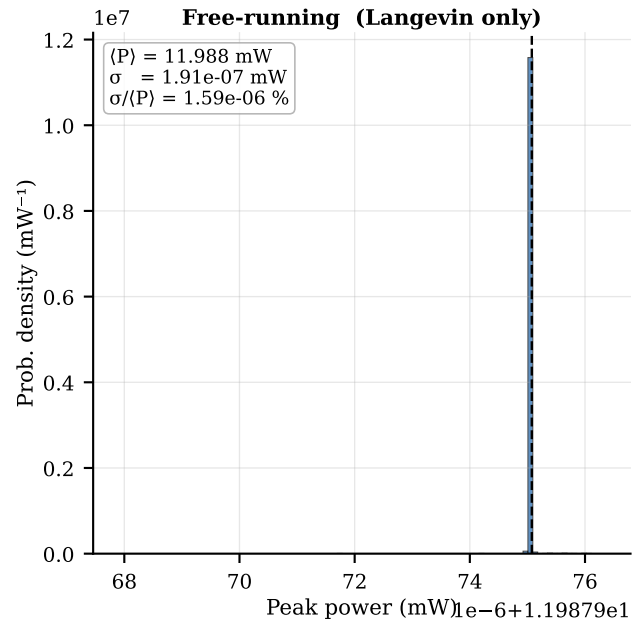
Repetition rate $f_{\text{rep}} = 6 \text{ GHz}$ ($T_{\text{rep}} = 0.17 \text{ ns}$) $N = 4,000$ pulses sampled at $t = 80.0 \text{ ps}$ into each pulse



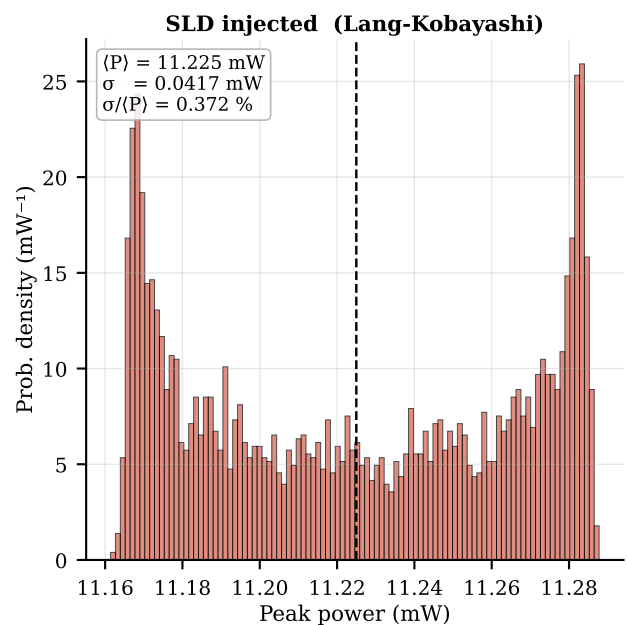
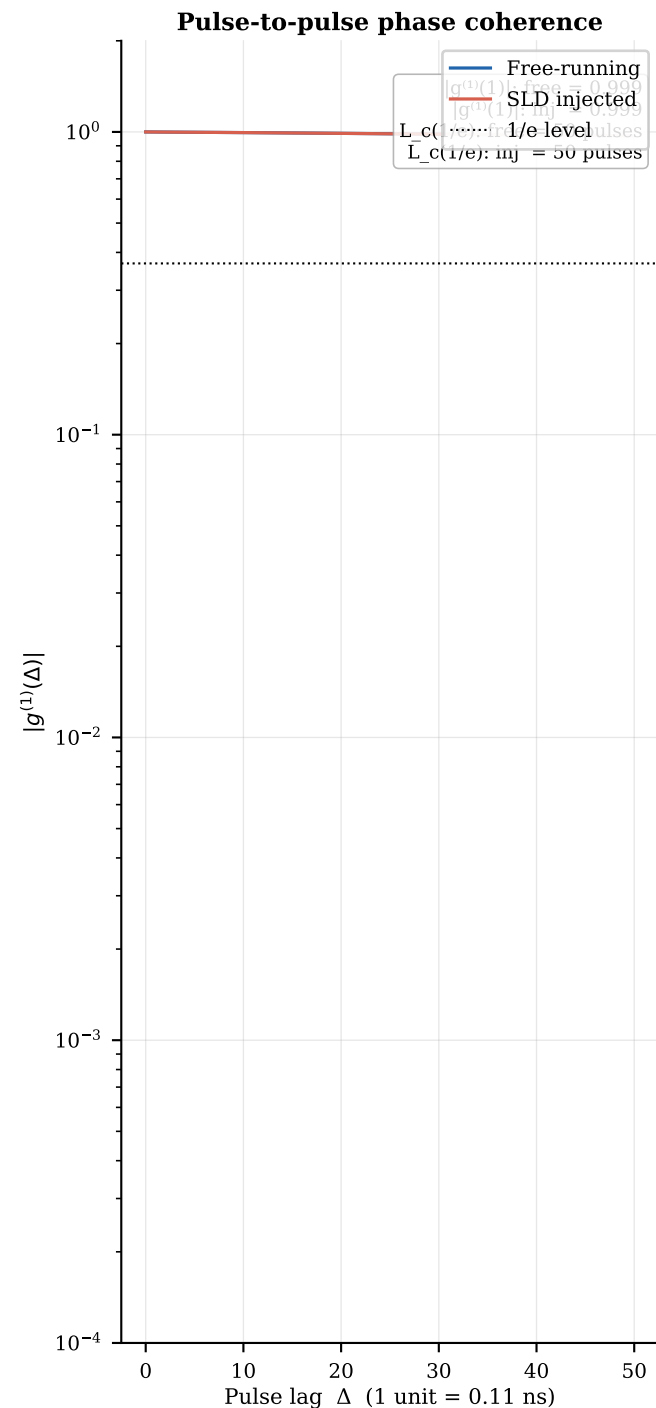
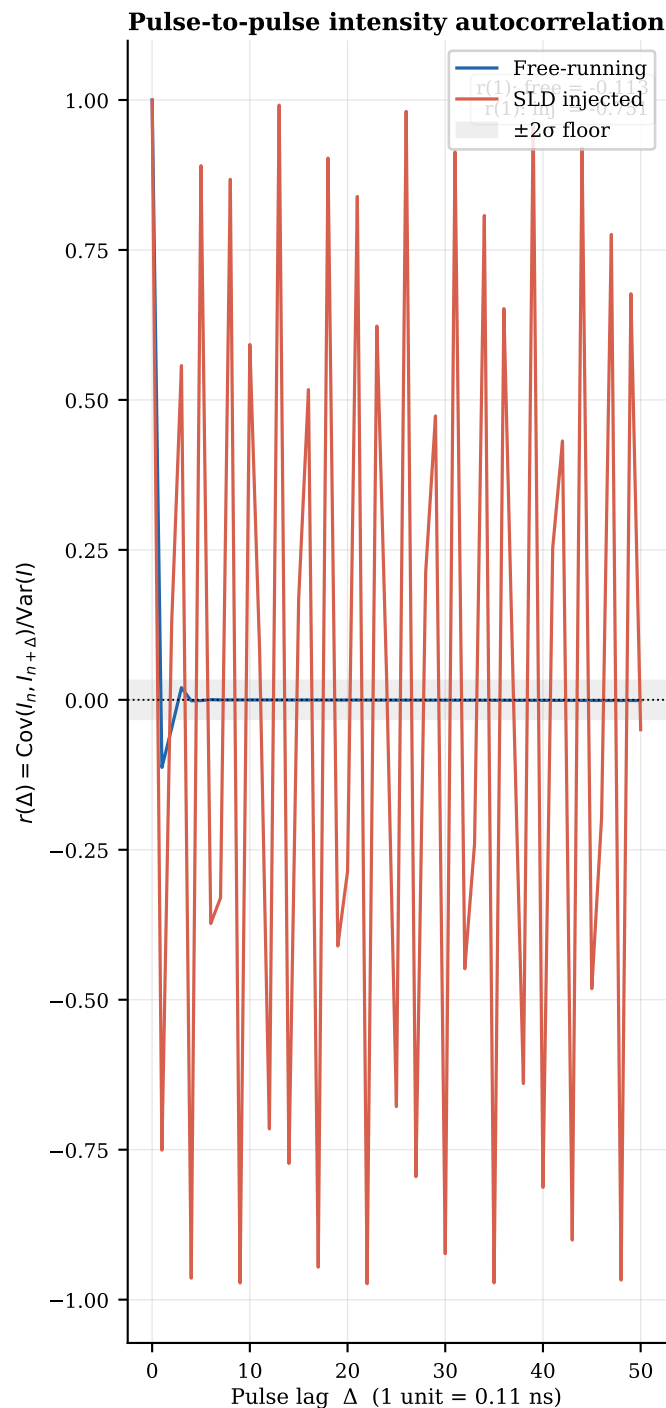
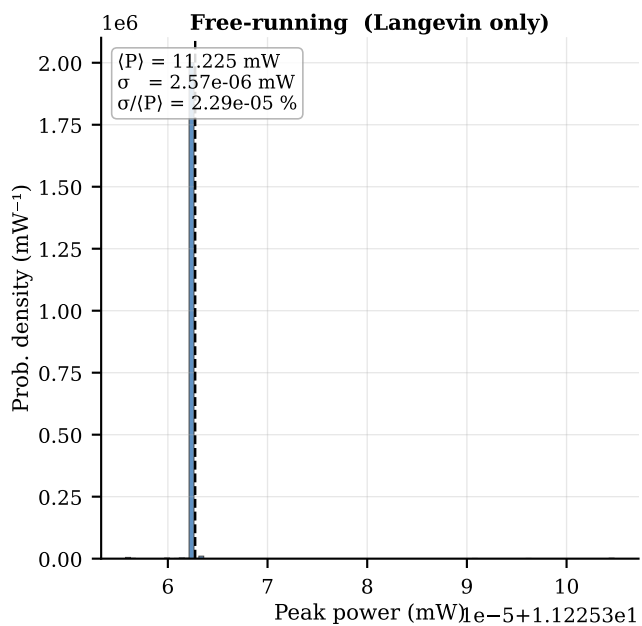
Repetition rate $f_{\text{rep}} = 7 \text{ GHz}$ ($T_{\text{rep}} = 0.14 \text{ ns}$) $N = 4,000$ pulses sampled at $t = 72.0 \text{ ps}$ into each pulse



Repetition rate $f_{\text{rep}} = 8 \text{ GHz}$ ($T_{\text{rep}} = 0.12 \text{ ns}$) $N = 4,000$ pulses sampled at $t = 66.0 \text{ ps}$ into each pulse

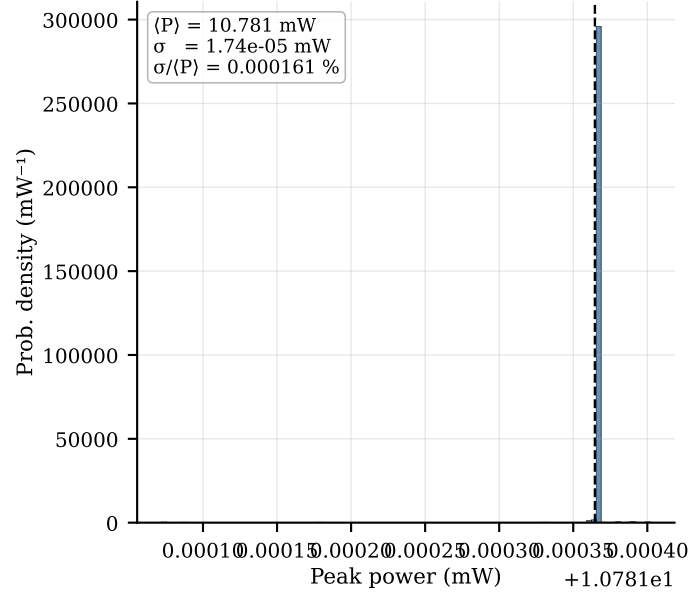


Repetition rate $f_{\text{rep}} = 9 \text{ GHz}$ ($T_{\text{rep}} = 0.11 \text{ ns}$) $N = 4,000$ pulses sampled at $t = 60.0 \text{ ps}$ into each pulse

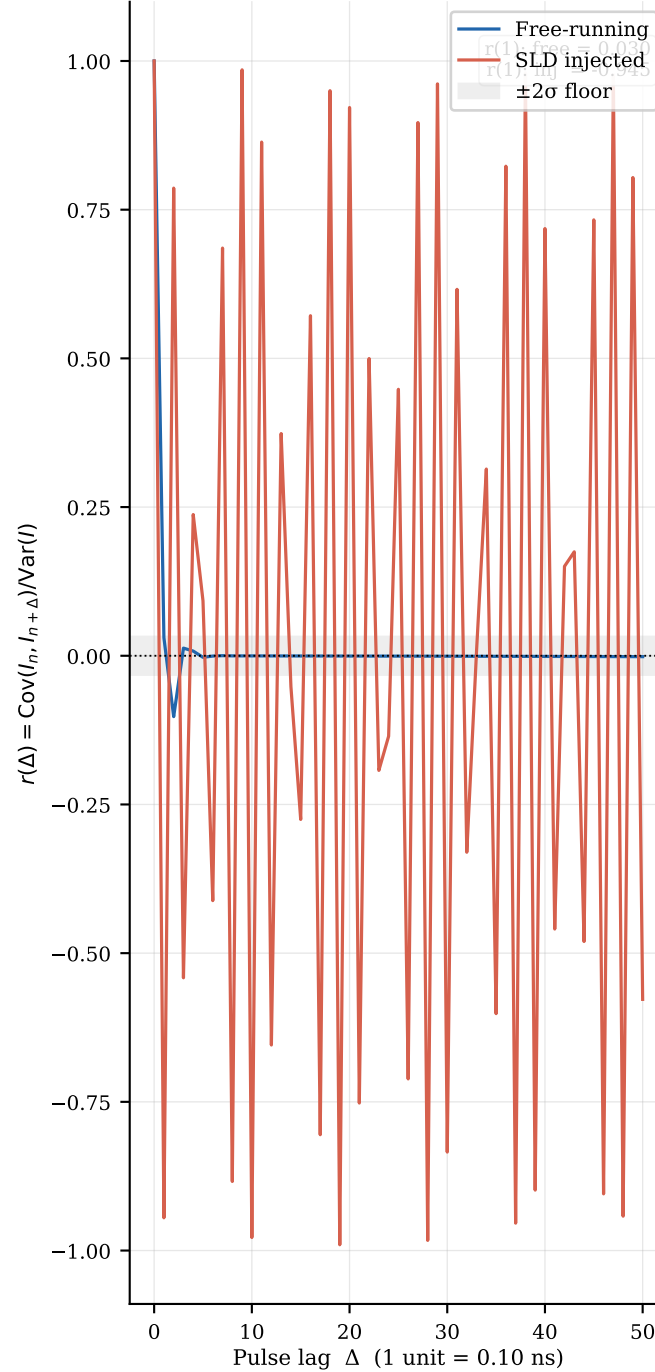


Repetition rate $f_{\text{rep}} = 10 \text{ GHz}$ ($T_{\text{rep}} = 0.10 \text{ ns}$) $N = 4,000$ pulses sampled at $t = 56.0 \text{ ps}$ into each pulse

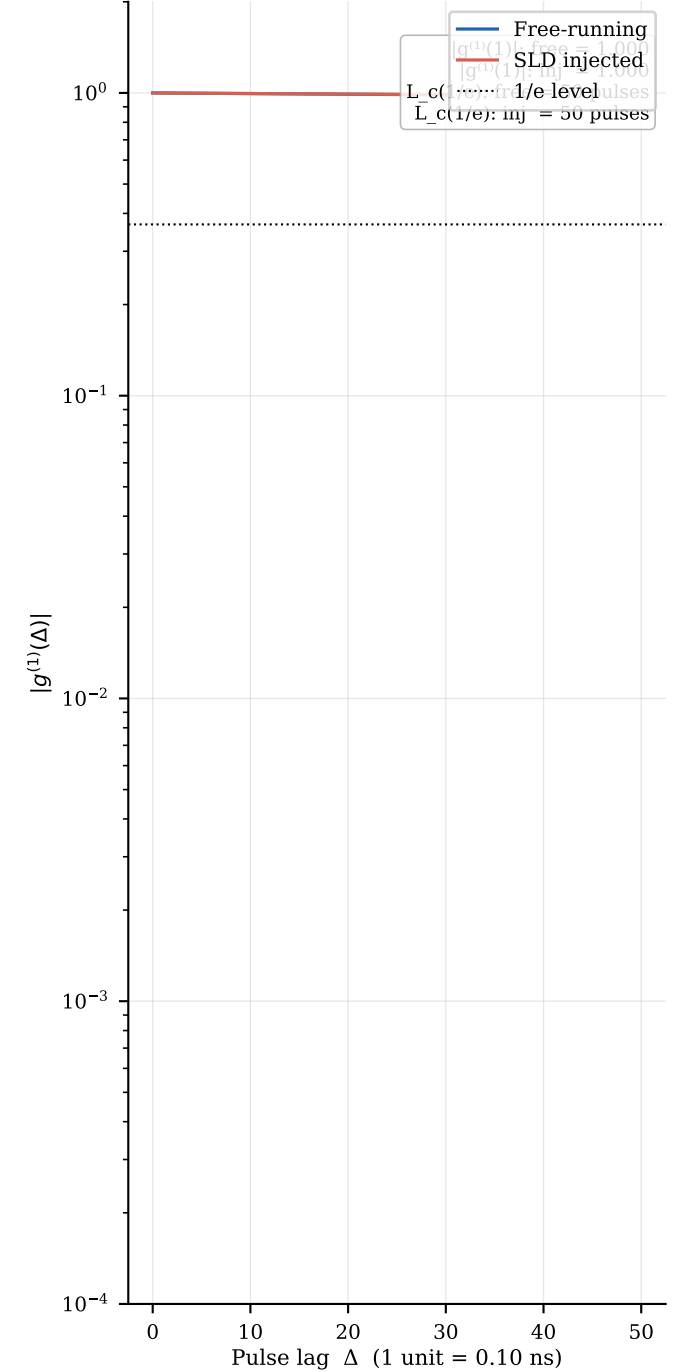
Free-running (Langevin only)



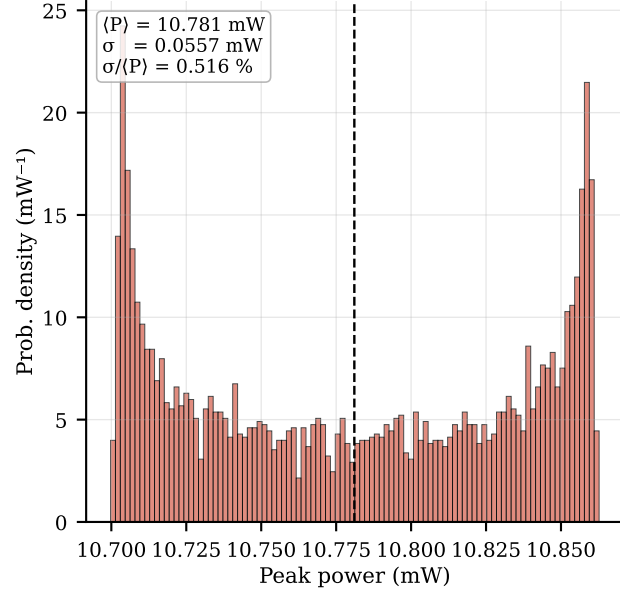
Pulse-to-pulse intensity autocorrelation



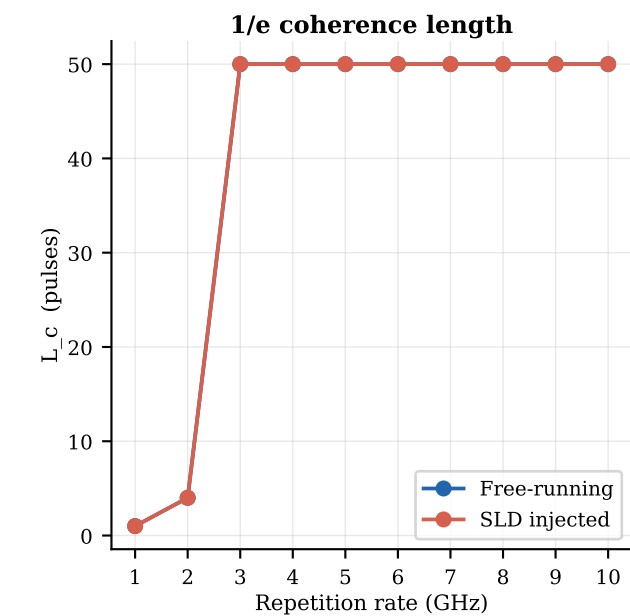
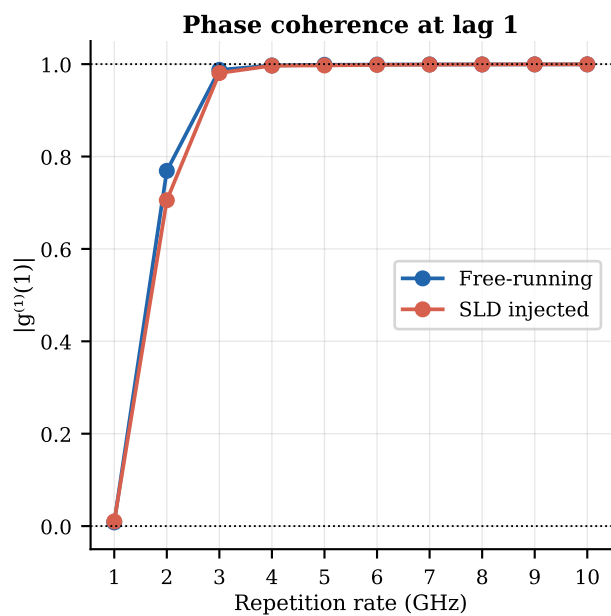
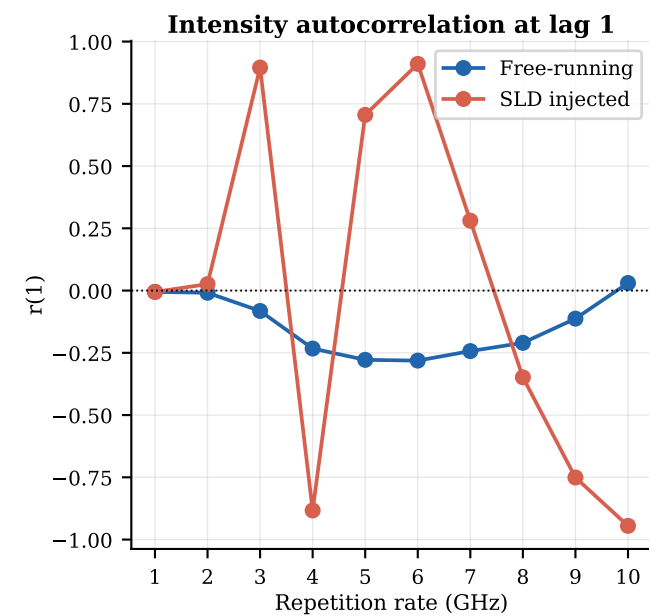
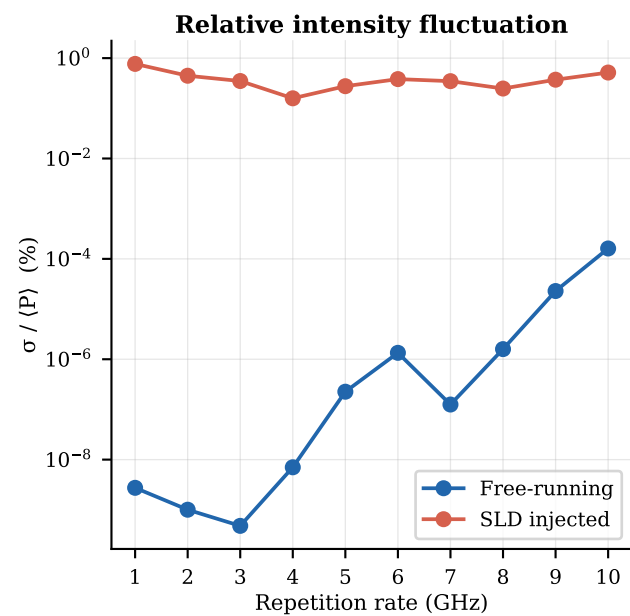
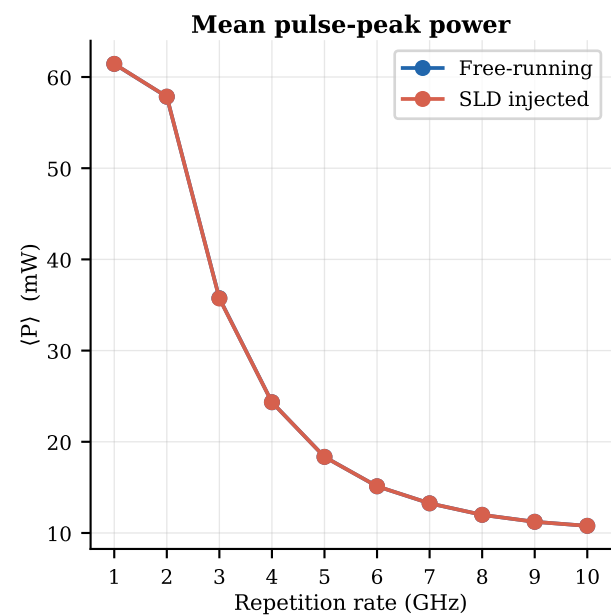
Pulse-to-pulse phase coherence



SLD injected (Lang-Kobayashi)



Summary: Pulse-to-Pulse Statistics vs Repetition Rate (1 - 10 GHz)



Phase coherence summary

f (GHz)	g [†] (1) free	g [†] (1) inj	L_c free	L_c inj
1	0.008	0.010	1	1
2	0.769	0.705	4	4
3	0.988	0.981	50	50
4	0.997	0.997	50	50
5	0.999	0.997	50	50
6	0.999	0.998	50	50
7	0.999	0.999	50	50
8	0.999	0.999	50	50
9	0.999	0.999	50	50
10	1.000	1.000	50	50